

The knight changes color every time

Chess Math

THE IDEA

Every knight move changes square color. An even number of knight moves returns to the original color; an odd number reaches the opposite color. This does not give the full shortest path, but it rules out impossible lengths.

WORKED EXAMPLE

A knight on b1 is on a light square. It cannot reach another light square in three moves. If your imagined three-move route ends on d5, check the route: d5 is light too.

TRY IT BEFORE READING ON

Can a knight move from a1 to b2 in two moves?

ANSWER AND REASONING

No. Although both squares are dark and an even move count is required, no square is a knight move from both a1 and b2. Parity is a necessary test, not a complete route finder.

TAKE THIS INTO YOUR NEXT GAME

Use color as a filter, then calculate the actual route.